Appendix for A Cost–Risk Joint-Oriented Learning framework for Wind Power Forecasting based on Metrics Learning

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1 Appendix

1.1 Computational Complexity

Let B, d, and p denote the batch size, forecasting-output dimension, and number of forecasting-model parameters, respectively. For objective $m \in \{c, r\}$, the metric-induced loss is

$$\mathcal{L}_m = \frac{1}{B} \sum_{b=1}^{B} \mathbf{e}_b^\top \mathbf{M}_m^\star \mathbf{e}_b, \quad \mathbf{e}_b = h_\theta(\mathbf{s}_b) - \widetilde{\mathbf{w}}_b. \tag{1}$$

For a dense Mahalanobis matrix $\mathbf{M}_m^\star \in \mathbb{R}^{d\times d}$, evaluating the product $\mathbf{M}_m^\star \mathbf{e}_b$ and its corresponding gradient $2\mathbf{M}_m^\star \mathbf{e}_b$ incurs a computational complexity of $\mathcal{O}(d^2)$ per sample. Consequently, optimizing the two metric objectives introduces an additional $\mathcal{O}(Bd^2)$ computational overhead and $\mathcal{O}(d^2)$ memory footprints per batch. Crucially, these metric matrices are learned offline and remain fixed throughout the training of the forecasting model.

To balance the two objectives, MGDA solves the following optimization problem:

$$\alpha^\star = \arg\min_{0\le\alpha\le1} \|\alpha \mathbf{g}_c + (1-\alpha)\mathbf{g}_r\|_2^2 \tag{2}$$

which admits the analytical closed-form solution:

$$\alpha^\star = \Pi_{[0,1]} \left(\frac{(\mathbf{g}_r - \mathbf{g}_c)^\top \mathbf{g}_r}{\|\mathbf{g}_c - \mathbf{g}_r\|_2^2} \right) \tag{3}$$

where $\Pi_{[0,1]}(\cdot)$ denotes the projection operator onto the interval $[0, 1]$. Since this step only involves the inner products of two p-dimensional gradient vectors, the computational cost of the MGDA step is strictly $\mathcal{O}(p)$.

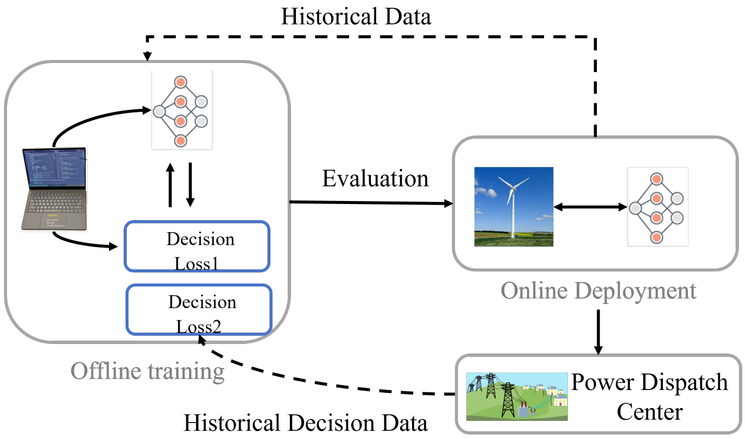

Figure 1: Flowchart of Model Deployment

Let C_f and C_b denote the computational costs of a single forward and backward pass, respectively. The overall per-batch training complexity is approximately:

$$\mathcal{O}\left(C_\mathrm{f} + 2C_\mathrm{b} + Bd^2 + p\right). \tag{4}$$

Although this is higher than standard single-loss training, both the Mahalanobis metric evaluation and the MGDA optimization are strictly confined to the offline training phase. During online deployment, only the optimized forecasting network h_θ is retained, thereby restoring the inference complexity to that of a standard forward pass, i.e., $\mathcal{O}(C_\mathrm{f})$.

1.2 Deployability

Upon completing the training phase, the metric-learning module, the MGDA solver, and the downstream optimization models are decoupled from the deployment pipeline. During online inference, the deployed system exclusively executes a forward pass(see Figure 1):

$$\overline{\mathbf{w}}_t = h_{\theta^\star}(\mathbf{s}_t) \tag{5}$$

thereby maintaining the identical architecture and computational footprint of a conventional forecasting model.

To adapt to non-stationary environments in real-world operations, the forecasting model can be periodically updated in a closed-loop manner using newly collected measurements and downstream decision-loss feedback:

$$\left\{\mathbf{M}_m^{(k+1)}, \theta^{(k+1)}\right\} = \mathcal{F}\left(\mathcal{D}^{(k)}, J_c^{(k)}, J_r^{(k)}\right) \tag{6}$$

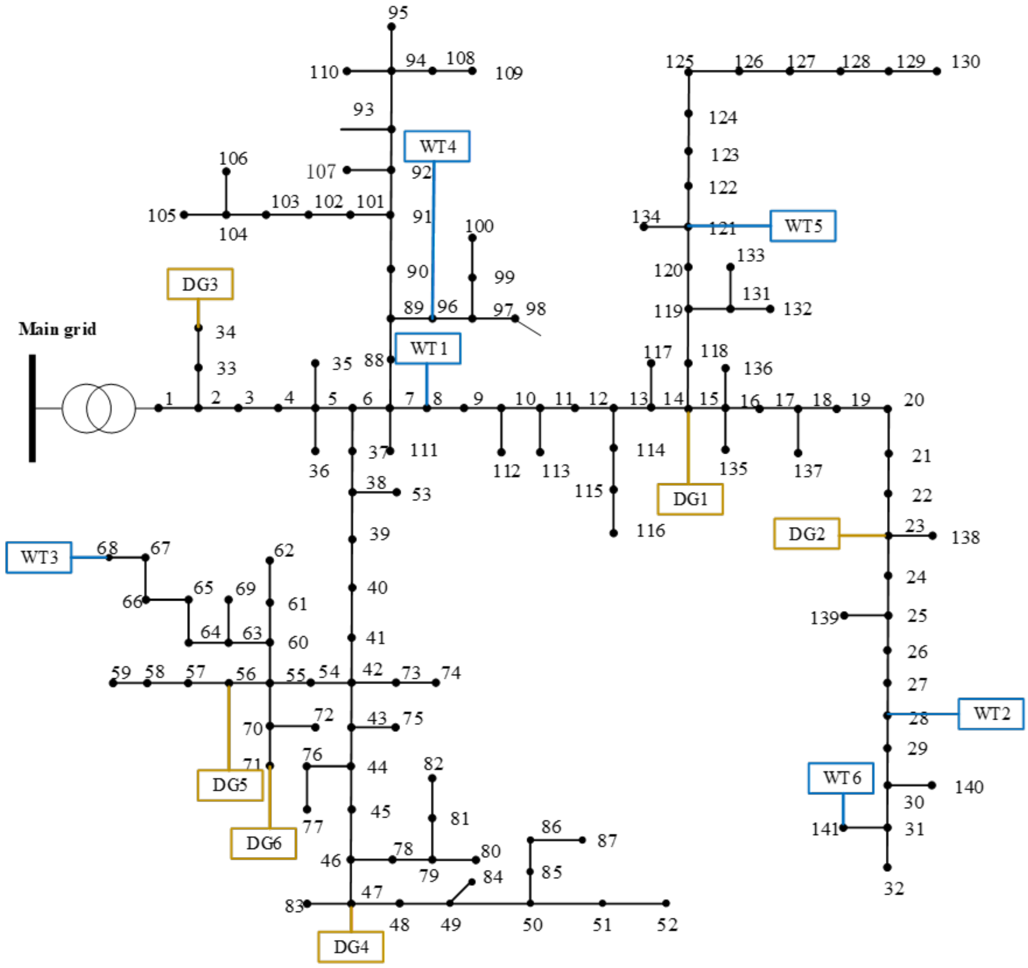

Figure 2: Modified IEEE 141-bus distribution network.

where $\mathcal{D}^{(k)}$ denotes the newly accumulated operational dataset at epoch k. Such coordination between renewable energy stations and dispatch centers establishes a novel decision-oriented model deployment paradigm, enabling forecasting models to be continuously calibrated against their downstream economic and risk impacts.

1.3 IEEE-141

To validate the scalability of the proposed method, tests are conducted on the IEEE 141-bus distribution network, as illustrated in Fig. 2.

1.4 Comparison with Implicit Differentiation

To isolate the effect of the gradient construction mechanism, we further consider a single-objective decision-focused forecasting problem that includes only the downstream dispatch-cost objective. The risk-oriented branch and MGDA coordination are removed in this experiment. Therefore, the downstream power-flow optimization is formulated as

$$\boldsymbol{u}^{\star}(\bar{\boldsymbol{w}}) = \arg\min_{\boldsymbol{u}} J_c(\boldsymbol{u}, \bar{\boldsymbol{w}}) \tag{7}$$

subject to

$$\boldsymbol{c}(\boldsymbol{u}, \bar{\boldsymbol{w}}) = \mathbf{0}, \qquad \boldsymbol{g}(\boldsymbol{u}, \bar{\boldsymbol{w}}) \leq \mathbf{0}, \tag{8}$$

where $\bar{\boldsymbol{w}} = h_\theta(\boldsymbol{s})$ denotes the predicted wind power, and $\boldsymbol{u}^{\star}$ contains the optimal dispatch variables, including grid purchase, DG outputs, wind-power utilization, nodal voltages, and branch power flows.

For implicit differentiation, the optimality conditions of the dispatch problem can be represented by the KKT residual

$$\boldsymbol{F}\left(\boldsymbol{z}^{\star}, \bar{\boldsymbol{w}}\right) = \mathbf{0}, \tag{9}$$

where

$$\boldsymbol{z}^{\star} = [\boldsymbol{u}^{\star}, \boldsymbol{\lambda}^{\star}, \boldsymbol{\nu}^{\star}] \tag{10}$$

collects the primal and dual variables. According to the implicit function theorem, the decision sensitivity is given by

$$\frac{\partial \boldsymbol{z}^{\star}}{\partial \bar{\boldsymbol{w}}} = -\left(\frac{\partial \boldsymbol{F}}{\partial \boldsymbol{z}}\right)^{-1} \frac{\partial \boldsymbol{F}}{\partial \bar{\boldsymbol{w}}}. \tag{11}$$

The forecasting-model gradient is then computed as

$$\nabla_\theta \mathcal{L}_c = \frac{\partial \mathcal{L}_c}{\partial \boldsymbol{u}^{\star}} \frac{\partial \boldsymbol{u}^{\star}}{\partial \bar{\boldsymbol{w}}} \frac{\partial h_\theta(\boldsymbol{s})}{\partial \theta}. \tag{12}$$

For the power-flow optimization considered in this work, implicit differentiation requires constructing and solving a KKT sensitivity system containing the power-balance equations, voltage constraints, branch-current limits, and generator-output constraints. Its dimension therefore increases with the numbers of buses, branches, dispatch variables, dual variables, and active constraints. Moreover, changes in wind-power forecasts may alter the active constraint set, which can further affect gradient stability.

In contrast, the proposed metric-based approach first learns the relationship between forecasting errors and dispatch-cost losses from offline samples:

$$\boldsymbol{M}_c^{\star} = \arg\min_{\boldsymbol{M} \succeq 0} \sum_{i=1}^{N} \left[\boldsymbol{e}_i^{\top} \boldsymbol{M} \boldsymbol{e}_i - \mathcal{L}_{c,i}\right]^2, \tag{13}$$

Table 1: Comparison of implicit and metric-based gradients

Method	E_C($)	Time (s)
Implicit-DFL	6318.34	6874.2
Metric-DFL	6321.16	291.6

where

$$\boldsymbol{e}_i = \bar{\boldsymbol{w}}_i - \tilde{\boldsymbol{w}}_i. \tag{14}$$

The resulting dispatch-oriented surrogate loss is

$$\widehat{\mathcal{L}}_c = \boldsymbol{e}^\top \boldsymbol{M}_c^\star \boldsymbol{e}, \tag{15}$$

and its gradient with respect to the forecasting-model parameters is explicitly obtained as

$$\nabla_\theta \widehat{\mathcal{L}}_c = 2\left(\boldsymbol{M}_c^\star \boldsymbol{e}\right)^\top \frac{\partial h_\theta(\boldsymbol{s})}{\partial \theta}. \tag{16}$$

Once $\boldsymbol{M}_c^\star$ is learned, each training update requires only a matrix–vector multiplication in the forecasting-error space. Therefore, the metric-based gradient no longer depends on the full set of primal variables, dual variables, and active constraints of the downstream power-flow problem. This replaces exact pointwise sensitivities with a reusable data-driven surrogate, thereby avoiding repeated construction and solution of the KKT sensitivity system during forecasting-model training.

To ensure a fair comparison, Implicit-DFL and Metric-DFL use the same forecasting backbone, training data, downstream dispatch model, initialization, and stopping criterion. The only difference lies in the computation of the decision-oriented gradient.

As shown in Table 1, the metric-based method achieves competitive dispatch performance while reducing the computational burden associated with repeated KKT differentiation. The advantage becomes more evident on the larger IEEE 141-bus system, where the KKT sensitivity system grows with the network variables and operational constraints, whereas the metric-induced gradient remains defined only in the forecasting-output space.